

Bluestock Fintech

Mutual Fund Analytics & Interactive Dashboard Development

Bluestock Fintech Internship – June 2026

Presented By
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Project Overview & Objectives



ETL Pipeline & Data Infrastructure

Build an end-to-end ETL pipeline to extract raw CSV data, clean duplicates, standardize dates, handle missing values, and load processed data into SQLite for structured analysis.



EDA & Risk Metrics

Perform exploratory data analysis and calculate advanced risk metrics including VaR, CVaR, Sharpe Ratio, CAGR, and volatility to evaluate fund performance and investor behavior.



Power BI Dashboard & Recommendations

Develop an interactive Power BI dashboard visualizing fund performance, investor demographics, and SIP trends, plus a fund recommendation system ranked by Sharpe Ratio.

Datasets Used

fund_master.csv

Contains mutual fund scheme details including fund names, categories, sub-categories, and AMFI codes. Serves as the master reference for all fund-level analysis.



nav_history.csv & investor_transactions.csv

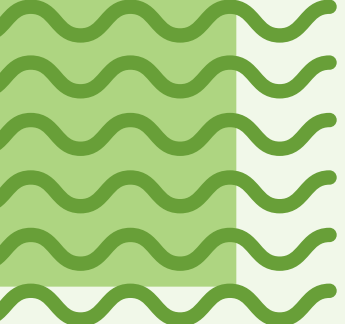
Historical NAV data tracks daily fund prices over time. Investor transactions capture SIPs, lumpsum investments, and redemptions with dates and amounts.



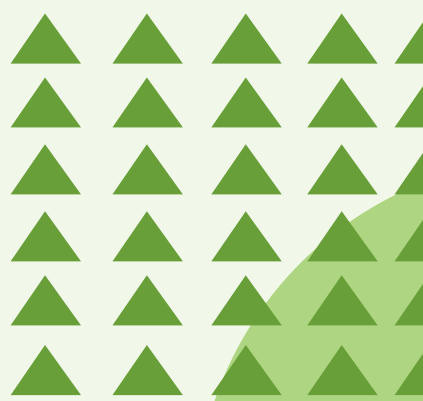
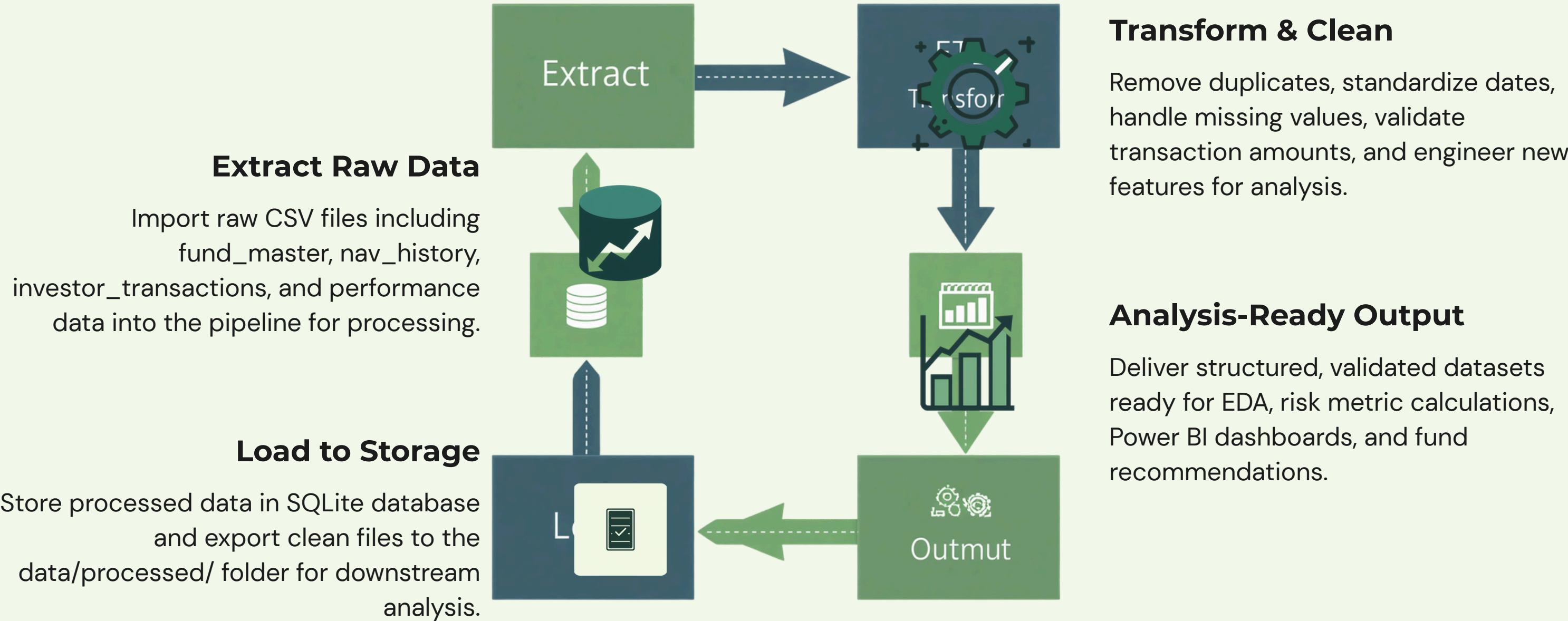
performance.csv

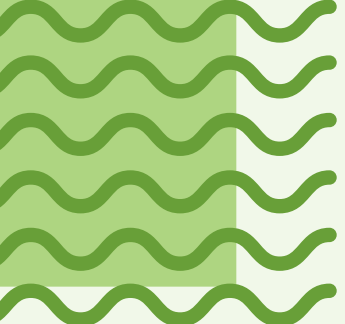
Stores computed metrics including returns, CAGR, Sharpe Ratio, volatility measures, and risk grades. Enables fund comparison and risk-adjusted performance analysis.





ETL Pipeline Overview





Exploratory Data Analysis – Key Insights



Fund & Transaction Distribution

Equity mutual funds dominate the dataset. SIPs constitute 58.4% of all transactions, followed by Lumpsum (35.34%) and Redemptions (6.17%), reflecting strong systematic investment preference.



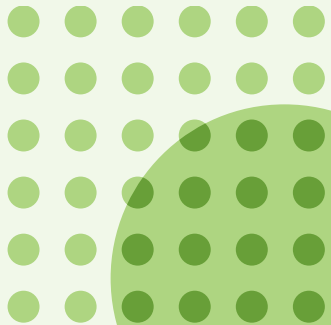
Investor Demographics

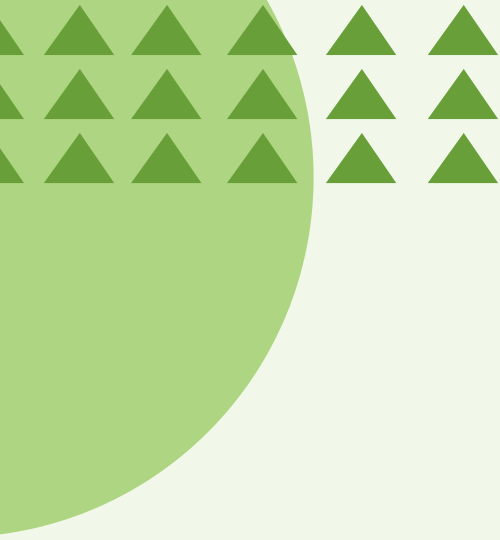
66.54% Male vs 33.46% Female investors. The largest age group is 26–35 years (41%), indicating strong participation among young working professionals.



Risk vs. Return & SIP Growth

Higher-risk funds yield significantly higher average returns (~22% vs ~6%). Monthly SIP inflows show a steady upward growth trend, signaling rising investor confidence.





Advanced Risk Metrics

VaR & CVaR Analysis



Historical VaR (95% Confidence)

Quantifies the maximum potential loss over a given period at a 95% confidence level. Helps set risk thresholds across fund categories.



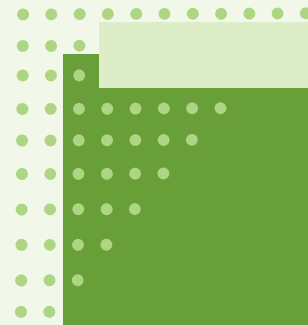
Conditional VaR (CVaR / ES)

Estimates the expected loss in worst-case scenarios beyond the VaR threshold. Provides a more conservative and complete risk picture.



Risk Management Application

VaR and CVaR are compared across fund categories and risk grades to identify high-exposure funds and guide portfolio risk decisions.



Rolling Sharpe Ratio Analysis



90-Day Rolling Methodology

Sharpe Ratio is computed on a rolling 90-day window to capture dynamic risk-adjusted returns, smoothing short-term noise while reflecting evolving fund performance.



Track Performance Fluctuations

Rolling analysis reveals how fund efficiency shifts over time, highlighting periods of outperformance and drawdown relative to risk taken by the fund manager.



Identify Consistent Outperformers

Funds maintaining a stable, high rolling Sharpe Ratio across market cycles are flagged as consistent outperformers and prioritized in the recommendation system.



Investor Cohort & SIP Continuity Analysis

Cohort Analysis by First Investment Year



Investors grouped by year of first investment to track retention, behavioral patterns, and long-term loyalty trends across cohorts.

SIP Continuity Flagging



SIPs with gaps exceeding 35 days are automatically flagged as at-risk, enabling proactive intervention to prevent lapse and improve continuity rates.

At-Risk Investor Detection



Identifies investors at risk of SIP lapse, quantifies loyalty metrics, and supports targeted retention strategies to reduce churn and improve AUM stability.



Power BI Dashboard – Fund Performance (Page 1)



Key Performance Metrics

Total Funds: 40 | Total AUM: 1 Million | Avg 3-Year Return: 14.09% | Avg Sharpe Ratio: 1.36. These headline KPIs provide an at-a-glance summary of the mutual fund portfolio's overall health and performance.



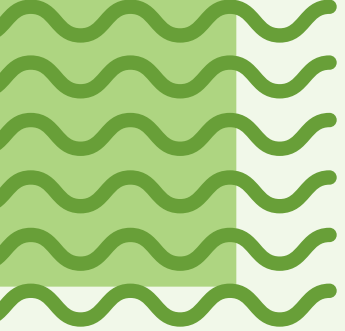
NAV Trends & Risk vs Return

Interactive NAV trend line chart tracks historical fund performance over time. Risk vs Return scatter plot maps each fund's annualized return against its volatility, enabling quick identification of high-performing, risk-efficient funds.



Gender & Investor Distribution

Gender distribution pie chart reveals 66.54% Male and 33.46% Female investors. Age cohort breakdown highlights the 26–35 segment as the dominant group at 41%, guiding targeted fund recommendations and communication strategies.



Power BI Dashboard – Industry Overview (Page 2)



Risk Grade Distribution

Moderate: 40%, High: 20%,
Low: 15%, Very High: 15%,
Moderately High: 10%. Higher-
risk funds average 3Y return
~22% vs lower-risk ~6%.



Top 10 Funds by 3- Year Return

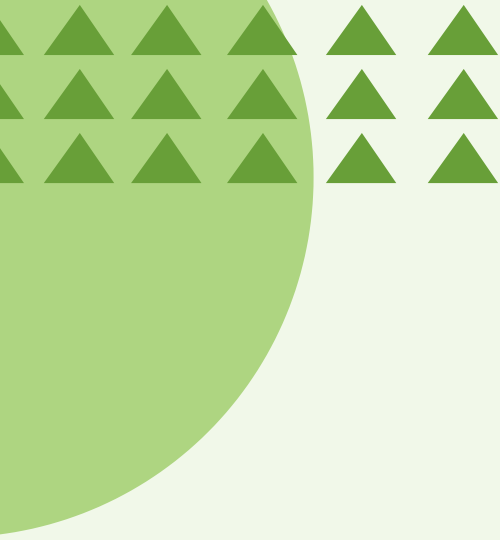
Ranked bar chart highlighting
top-performing mutual funds.
Leading funds deliver 3-year
returns exceeding 22%, driven
by equity mid-cap and small-
cap categories.



Fund Distribution by Category

Equity funds: 39K
transactions, Debt funds: 7K.
Age group and state-wise
investment breakdowns reveal
regional preferences and
demographic trends.





Power BI Dashboard

Investor Analytics (Page 3)



Transaction Type Distribution

SIP: 58.4% of all transactions.
Lumpsum: 35.34%.
Redemption: 6.17%. SIPs dominate, reflecting strong systematic investment behavior among investors.



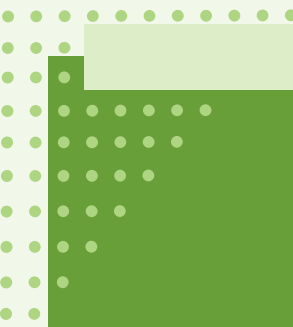
State & Age Group Analysis

Investment volumes mapped by state and age group. Largest investor segment: 26–35 years (41%). Geographic spread reveals top contributing states and regional trends.

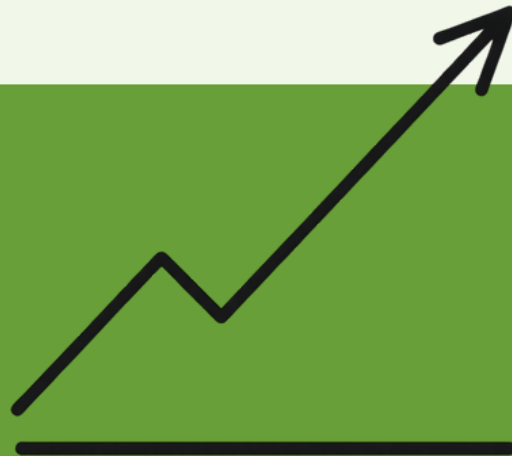


Gender & City Tier Breakdown

66.54% Male, 33.46% Female investors. City tier analysis compares Top 30 vs Bottom 30 cities, highlighting urban concentration and growth potential in Tier-2 markets.

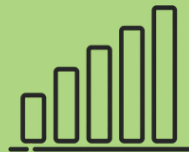


Power BI Dashboard – SIP Market Trends (Page 4)



Monthly SIP Inflow Trends

Line chart tracks monthly SIP inflows showing steady growth over time, highlighting seasonal patterns and acceleration in retail investor participation across fund categories.



Risk Grade & Category Breakdown

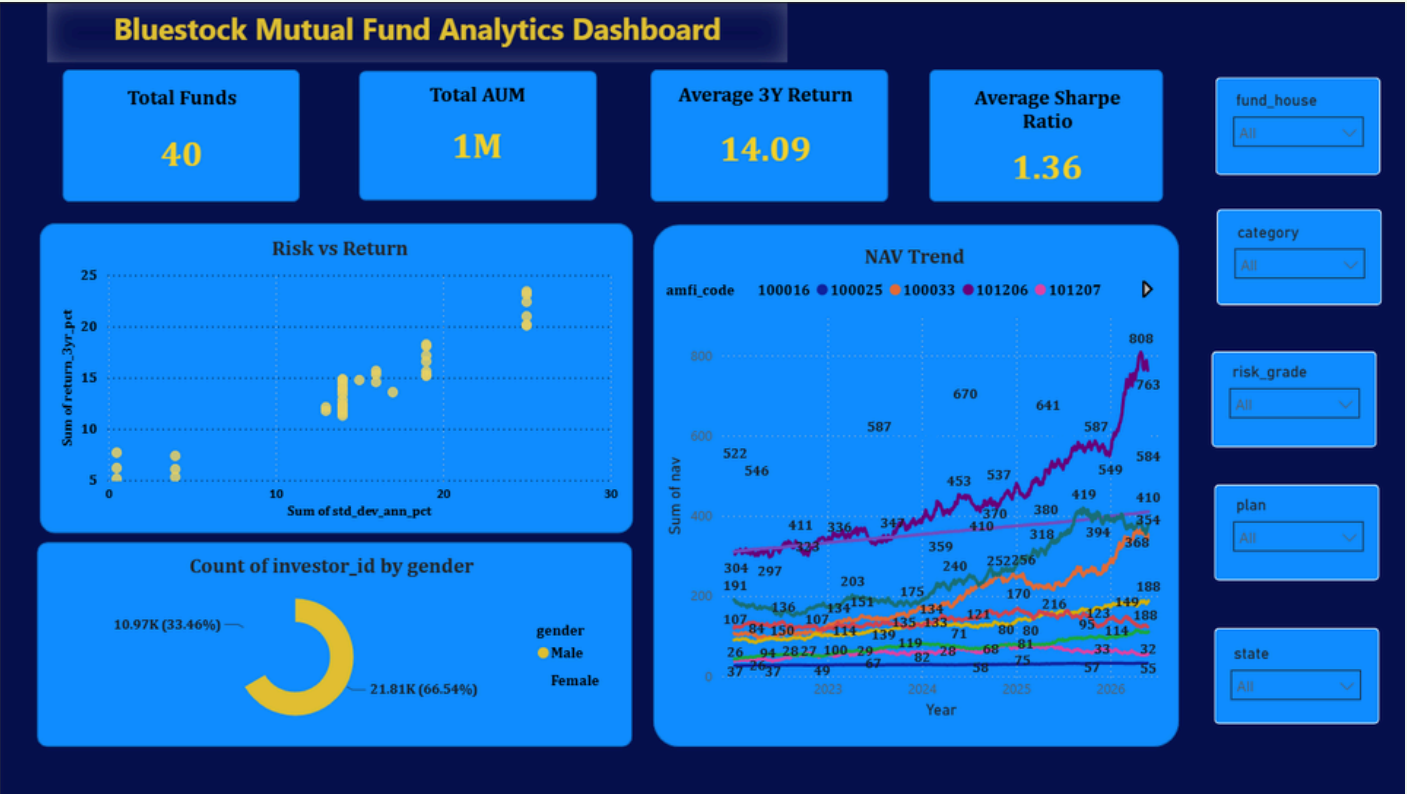
Bar charts display risk grade distribution and investment by fund category — Equity leads at 39K, followed by Debt at 7K, with Moderate risk grade at 40% of total funds.



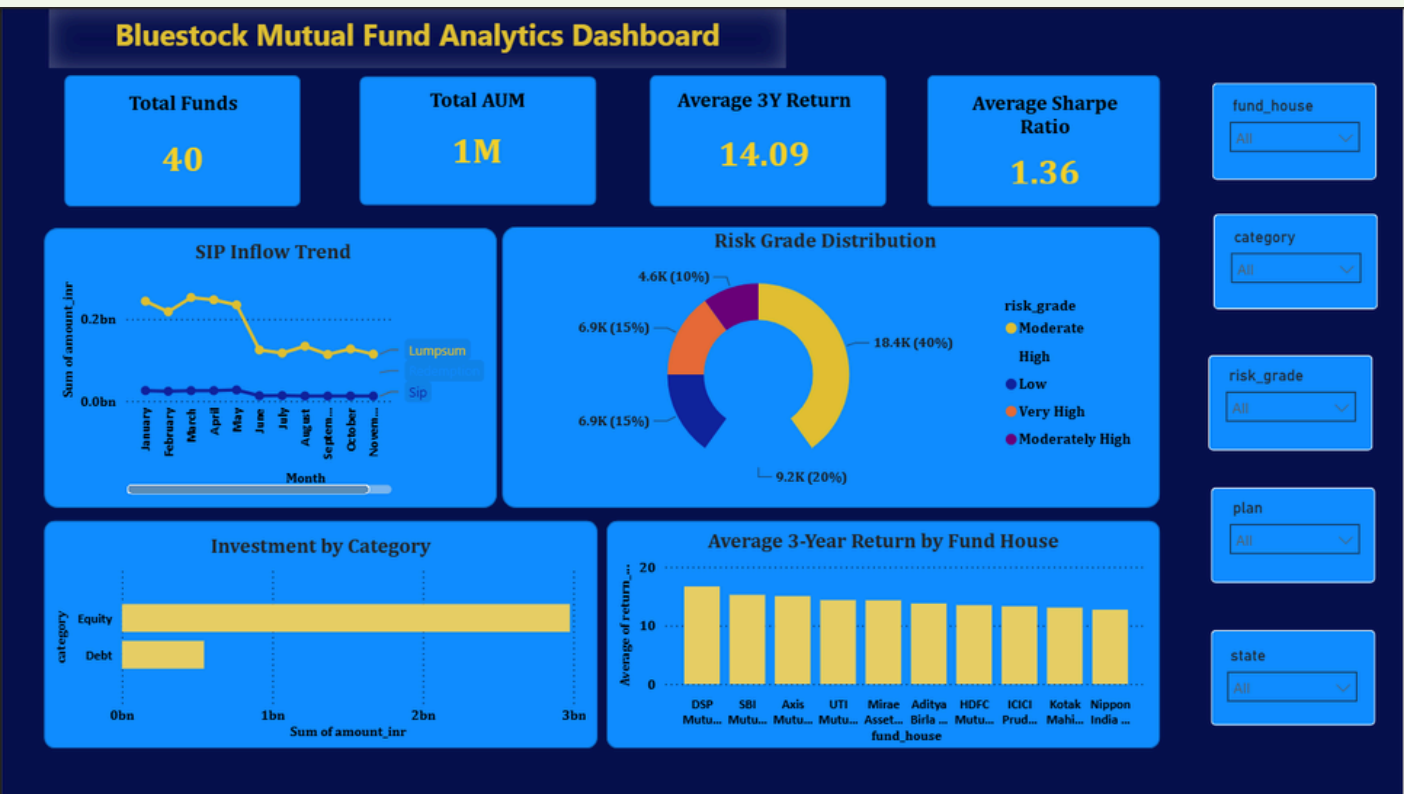
Avg 3-Year Return by Fund House

Ranked bar chart compares average 3-year returns across fund houses. High-risk fund houses average ~22% returns vs ~6% for lower-risk peers, guiding investor allocation decisions.

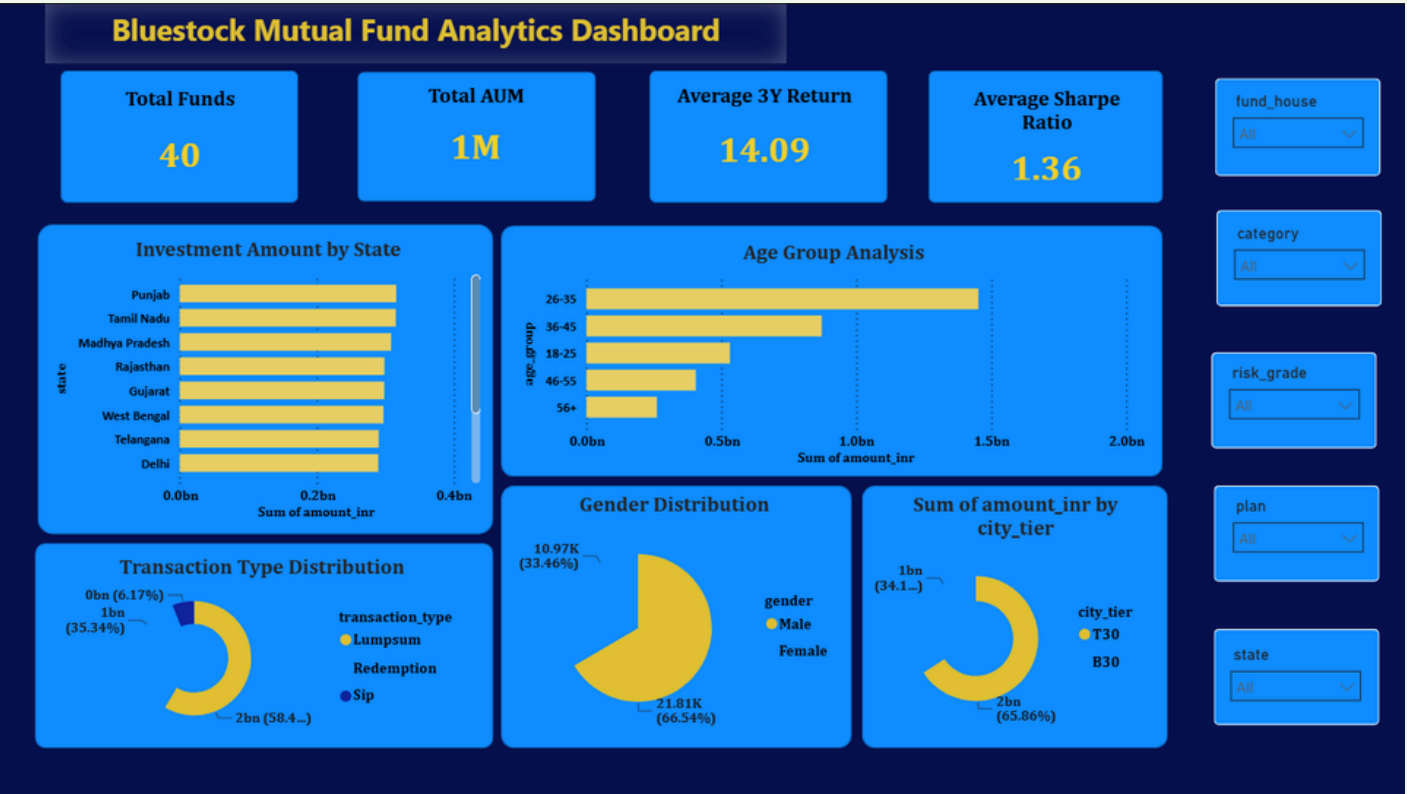
Fund Performance Overview (Page 1)



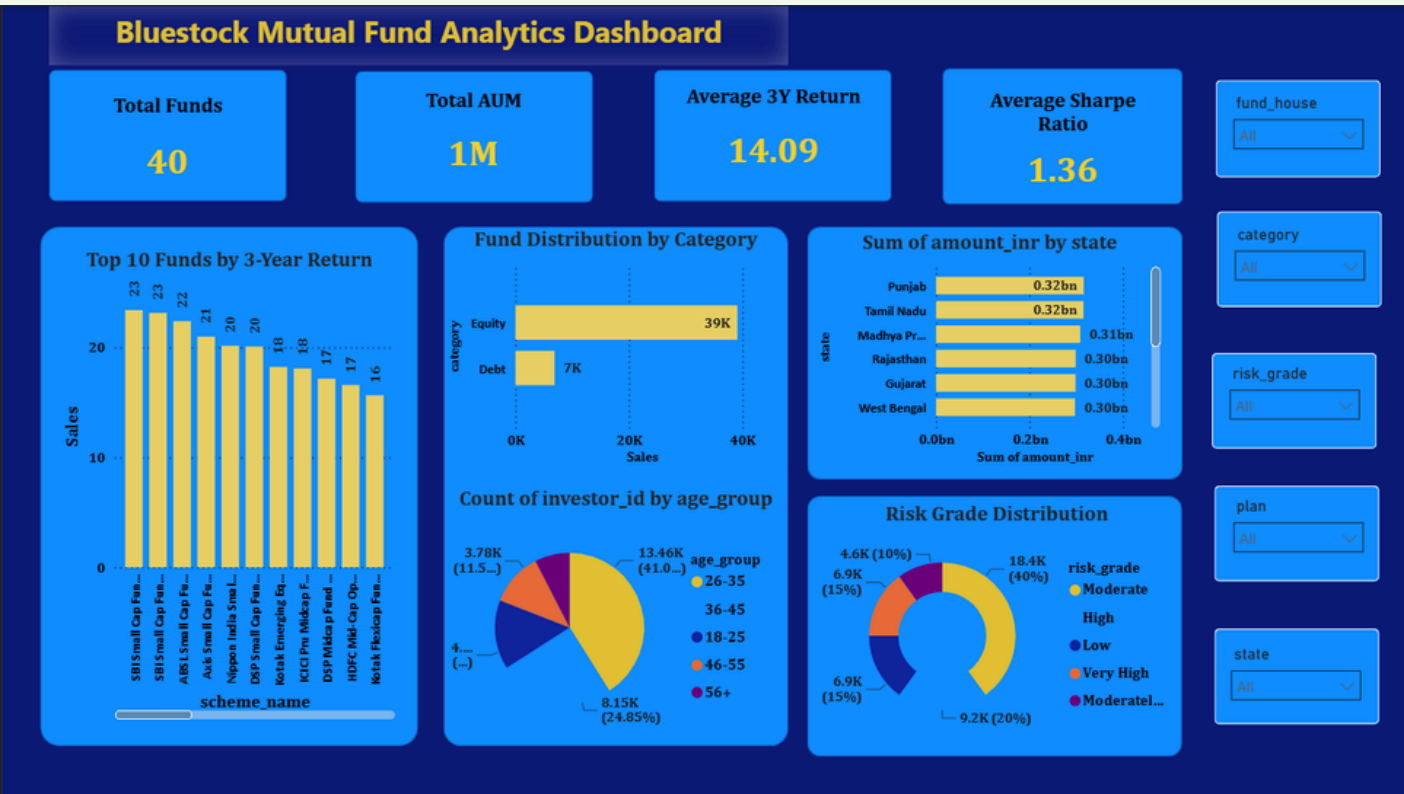
Industry & Category Analysis (Page 2)



Investor Analytics (Page 3)



SIP Market Trends (Page 4)



Key Metrics & Risk Analysis Summary



Investor Demographics

- Total Investors: 32,780
- Male: 21,810 (66.54%)
 - Female: 10,970 (33.46%)
- Transactions:
- SIP: 58.4%
 - Lumpsum: 35.34%
 - Redemption: 6.17%



Risk Grade Distribution

- Moderate: 40%
 - High: 20%
 - Low: 15%
 - Very High: 15%
 - Moderately High: 10%
- Higher-risk funds avg 3Y return
~22% vs lower-risk ~6%



Top Sharpe Ratio Performers

- Best risk-adjusted returns:
1. ICICI Pru Liquid Fund
 2. Kotak Liquid Fund
 3. ABSL Liquid Fund
- Avg Portfolio Sharpe Ratio: 1.36



Mutual Fund Recommendation System

Step 1: Filter by Risk Appetite



Funds are filtered based on the investor's risk grade preference (Low, Moderate, High, Very High) to ensure suitability.

Step 2: Rank by Sharpe Ratio



Within each risk category, funds are ranked by Sharpe Ratio to identify the best risk-adjusted return performers.

Step 3: Top 3 Recommendations



High Risk Example:

- Kotak Emerging Equity (0.96)
- ICICI Pru Midcap (0.95)
- DSP Midcap Fund (0.90)



Future Scope & Closing



Future Enhancements

Live NAV data integration via APIs, ML models for return prediction, and personalized recommendations using collaborative filtering.



Real-Time Deployment & Automation

Real-time Power BI dashboard deployment with automated data refresh pipelines for continuous, up-to-date fund analytics.



Thank You for Your Attention

Open for questions and discussion. Presented by Divyansh Dave – Bluestock Fintech Internship, June 2026.